Flag completion and Wilson symmetry in the four-type construction

Human Compression project: L023 independent Section 5 recovery

This paper concerns Construction 6, Lemmas 7–8 and Theorem 10 of Leemans–Stokes, arXiv:2208.08215v2, physical pages 10–12. The source identifies two notions of map: a faithful vertex–edge–face incidence map and an arbitrary quotient of the universal map group. They need not retain the same incidence information. Our counterexamples below are actual finite, faithful, thin incidence maps coming from surface embeddings; they therefore do not depend on that ambiguity. The conclusion about trialities without dualities survives for their four-type pregeometries, but the asserted universal flag completion does not.

1 Local chambers and the precise reconstruction hypothesis

Write a, b, c for a marked triple of nonidentity involutions, with $ac = ca$ and $V = \langle a, c\rangle$ of order four. Set

$$H_0 = \langle a, b\rangle, \quad H_1 = V, \quad H_2 = \langle b, c\rangle, \quad H_3 = \langle b, ac\rangle.$$

Objects of type i are left cosets of H_i; incidence means nonempty intersection. Types $0, 1, 2, 3$ are respectively faces, edges, vertices and Petrie polygons. We distinguish a full-type chamber from a maximal flag which might fail to have every type.

Lemma 1. *Suppose $G = \langle a, b, c\rangle$ and each of H_0, H_2, H_3 is proper. Every flag containing an edge extends to a full-type chamber. There are exactly two G-orbits of such chambers, both free; if G is finite their total size is $2|G|$. The pregeometry is a geometry if and only if every mutually incident face–vertex–Petrie triple has a common incident edge.*

Proof. If H_0 contained c it would be all of G, so $H_0 \cap V = \langle a\rangle$. Similarly $H_2 \cap V = \langle c\rangle$ and $H_3 \cap V = \langle ac\rangle$. The objects of each nonedge type incident with the edge V are its two cosets modulo the indicated subgroup. Any two such objects of different types meet inside V: two affine lines of different directions in C_2^2 meet. Thus the residue of V is a complete three-partite graph with two objects of each type. It has eight chambers, and every one of its flags extends. Translation by V acts freely on these eight choices (the three line kernels have trivial intersection), with two orbits. Since G is edge-transitive, these give exactly two free global orbits. In the finite case there are $8(|G|/4) = 2|G|$ chambers.

Every pair of incident nonedge cosets has an actual common group element g, so the edge gV extends that pair. Singletons also extend. The only remaining possible obstruction is therefore a three-type flag avoiding edges, exactly as stated. □

The last condition is a genuine global incidence condition. It does not follow merely from the description of a local Petrie turn. If the starting rank-three truncation is the faithful thin incidence map M, it reconstructs M literally: deleting type 3 preserves all its objects and incidences. In a quotient model with repeated corners forgotten, this must instead be checked; it is not supplied by group generation alone.

Theorem 2. *Let M be a reflexible Class III map whose rank-three coset truncation is its faithful thin incidence map. Then the four-type pregeometry has trialities and no type-transposing*

correlations. Its type-permutation quotient is exactly C_3. Its full-type chambers have the two orbits of Lemma 1. It is an incidence geometry only if the additional completion condition in that lemma holds.

Proof. A marked triality cycles a, c, ac and fixes b, so the three periods are equal, say m, and the three nonedge parabolics have order $2m$. None can be G: otherwise they all equal the same dihedral group. Infinite dihedral groups have no subgroup V of order four. If its rotation subgroup has even order and $m > 2$, its central involution is characteristic and lies in V, contradicting its cyclic permutation; if its rotation order is odd it has no V. For $m = 2$ the generators commute, and the existence of the marked triality forces $b \notin V$. Then $G = C_2^3$, whose full S_3 on $V \setminus 1$ fixes b, giving Class IV. The case $m = 1$ makes $b = a$ and contradicts its being fixed while a is cycled. Thus $m > 2$, and Lemma 1 applies.

An edge is incident with exactly two objects of each other type. An object of any nonedge type is incident with m edges, since $[H_i : H_i \cap V] = m$ (an infinite cardinal if the period is infinite). Since $m > 2$, the edge type is the unique type whose objects have exactly two neighbors of every other type. A correlation therefore fixes the edge type, without needing finite cardinalities. The marked triality induces an actual order-three coset permutation cycling the other three types. If any correlation induced a transposition on those types, conjugating by this triality would give one which swaps faces and vertices while fixing the Petrie type. Its restriction to the faithful rank-three truncation would be a duality of M. By flag transitivity, compose it with a map automorphism so that it fixes a chosen flag as a set while swapping its endpoint types. Unique flag adjacencies then give a marked automorphism interchanging a, c and fixing b. This is a Wilson duality, contrary to Class III. The image on types is consequently the displayed C_3. This argument uses neither flag completion in rank four nor an unsupported equality $\mathrm{Aut}(\Gamma) = G$. □

2 A faithful toroidal obstruction to flag completion

Take the square tessellation modulo the translations $(2, 0), (0, 2)$. Write its vertices v_{ij}, horizontal edges h_{ij} from (i, j) to $(i + 1, j)$, vertical edges e_{ij} from (i, j) to $(i, j + 1)$, and faces f_{ij}, with all indices modulo two. Parallel edges with the same endpoints remain distinct. A face has boundary

$$h_{ij},\ e_{i+1,j},\ h_{i,j+1},\ e_{ij}.$$

This is a two-cell torus map with four vertices, eight edges and four square faces. Every incident vertex–face pair has two boundary edges; every edge has two endpoints and two faces. Its 32 flags are therefore faithful incidence triples, with unique flag adjacencies.

Let D_8 be the group of eight integral signed permutation matrices in two coordinates and put $T = \mathbb{F}_2^2 \rtimes D_8$. Crucially, we retain the eight matrices as distinct: their action on the four vertices alone is not faithful. Multiplication is $(t, A)(u, B) = (t + Au, AB)$, with translations reduced modulo two. Use

$$a = ((1, 0), \mathrm{diag}(-1, 1)), \quad b = (0, \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)), \quad c = (0, \mathrm{diag}(1, -1)).$$

These are exactly the three reflections at the flag (v_{00}, h_{00}, f_{00}). They generate T: b, c generate D_8, *abcb* is translation $(1, 0)$, and conjugating that translation by b gives $(0, 1)$. Their action on the 32 flags is regular. Conversely an incidence automorphism fixing one flag fixes all flags by unique adjacencies and connectedness, so the full incidence automorphism group is exactly T.

Proposition 3. *The four-type coset pregeometry of this reflexible map is not a geometry. In fact (f_{00}, v_{11}, P) is an unextendable flag, where*

$$P = (h_{00}, e_{11}, h_{11}, e_{00})$$

is a Petrie circuit.

Proof. The displayed circuit alternately turns through the two sides of the square tessellation, so is a Petrie polygon. It contains all four vertices. The face f_{00} and P have precisely the edges h_{00}, e_{00} in common, meeting at v_{00}. Hence they are incident under both the literal coset definition and the source's two-edge description. The opposite vertex v_{11} belongs to both polygons but to neither of those two common edges. Thus the three objects are pairwise incident, while no edge completes them. This is an actual surface-map counterexample to source Lemma 7, not merely a degenerate quotient of its abstract map group. □

We will also need the Wilson symmetries and a precise word version of this witness. Set $t = ((1,1), I)$ and $j = (0, -I)$ in T. Direct multiplication gives

$$(ab)^2 = tj, \qquad (bc)^2 = j, \qquad (acb)^2 = t. \tag{1}$$

The first two squares are independent central involutions. The presentation by a, b, c as involutions, $[a, c] = 1$, with $[a, b]$ and $[b, c]$ central, has at most 32 elements: commute letters to the form $a^A c^C b^B [a, b]^X [b, c]^Y$, with five binary exponents: central commutators of two involutions have order at most two. Its actual image T has 32 elements, so this is an exact presentation. Every invertible linear change in $\langle a, c\rangle$ fixing b preserves it. In particular

$$a \mapsto c, \quad c \mapsto ac, \quad b \mapsto b$$

is a triality, and swapping a, c is a duality. This torus map is Class IV; it is not by itself a counterexample involving Class III.

For the coset version of Proposition 3, take $F = H_0$, $Q = tH_2$, $P = H_3$. Here $t \in H_3$ and $t = (ab)^2(bc)^2 \in H_0 H_2$, so all three pairs meet. The geometric argument proves that no coset of V meets all three.

3 The obstruction persists for an actual Class III map

Theorem 4. *Let S be a finite perfect group with no nontrivial normal 2-subgroup, supporting a faithful regular Class III map with markers a_0, b_0, c_0 and common odd period $m > 2$. Then the parallel product with the torus above is a faithful regular Class III map. Its four-type coset pregeometry is not a geometry.*

Proof. In $S \times T$ use the diagonal markers $A = (a_0, a)$, $B = (b_0, b)$, $C = (c_0, c)$. They generate a subdirect product. A proper subdirect product would give a nontrivial common quotient of S and T: project the two projection kernels and identify the quotient using any lift. Such a quotient would be both perfect and a finite 2-group, which is impossible. Thus the marked group is $G = S \times T$.

For each nonedge parabolic the rotation in S has order m and that in T has order four. Since these orders are coprime, its cyclic rotation group is their full direct product. The entire parabolic is therefore precisely the pairs of dihedral elements with equal rotation/reflection parity, and has order $8m$. The two original end parabolics in each factor intersect in its marked b subgroup. Their fiber products consequently intersect in $\langle B\rangle$. They each intersect $\langle A, C\rangle$ in the appropriate marked endpoint. This is the rank-three intersection property: the resulting coset map is thin with faithful flags. More explicitly, each edge has two faces and two vertices, and its four choices are a single free $\langle A, C\rangle$-orbit. An incident face–vertex pair has exactly two edges because the end-parabolic intersection has order two; every such pair extends by choosing an actual common group element. Thus the flags are exactly the regular G-orbit, and their connected adjacency graph is the Cayley graph of the generators. Gluing its flag triangles yields a closed surface map, with dihedral face and vertex residues. The same free-flag argument as above identifies its full incidence automorphism group with G.

The two marked trialities act componentwise. If the product map had a Wilson duality, the corresponding marked automorphism of G would descend through the characteristic subgroup $O_2(G) = T$ to a Wilson duality of the original S map. The equality follows from the hypothesis $O_2(S) = 1$. This is impossible, so the product has Class III. Its common period is $4m$.

Let K_i denote its four parabolics, and put $z = (1, t)$. Since m is odd, (1) and the order-m rotations give

$$(ACB)^{2m} = z, \qquad (AB)^{2m} = (1, tj), \qquad (BC)^{2m} = (1, j).$$

Thus $z \in K_3 \cap K_0K_2$ and (K_0, zK_2, K_3) is a pairwise incident flag. Projection to T maps each coset of K_i onto the corresponding coset of H_i and preserves incidence. An edge completing this flag would therefore complete the impossible torus flag (H_0, tH_2, H_3). There is none. □

An explicit instance is available without a large enumeration. Work in $\mathbb{F}_8 = \mathbb{F}_2[u]/(u^3+u+1)$ and use matrices

$$a_0 = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}, \quad b_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad c_0 = \begin{pmatrix} 1 & u^2 \\ 0 & 1 \end{pmatrix}.$$

The complete small certificate accompanying this paper constructs all 504 determinant-one matrices, verifies that these three generate all of them, checks the three periods nine and the end intersection $\langle b_0 \rangle$, and checks the field-squaring triality. Its finite execution is a declared premise for this fixed instance, not an exhaustion of maps. To exclude a marked duality it evaluates the literal word

$$w = ababababcbcbcbababc.$$

The word is the identity on (a_0, b_0, c_0); interchanging a_0, c_0 gives $\left(\begin{smallmatrix} u^2+u & u^2+u \\ u & 1 \end{smallmatrix}\right)$, which is not the identity. Triality then excludes the other two Wilson transpositions as well.

For completeness the other group hypotheses have short proofs. The group $\mathrm{SL}_2(8)$ is perfect: for a diagonal matrix with diagonal entries d, d^{-1}, $d \neq 1$, commutation with upper unipotents gives every upper unipotent since $d^2 - 1 \neq 0$; lower unipotents are conjugate, and elementary elimination shows these generate the whole determinant-one group. Every involution is a non-identity unipotent and these form one conjugacy class. Its centralizer has order eight, as direct two-by-two commutation shows, so the class has 63 elements. A nontrivial normal 2-subgroup would contain an involution and hence all 63, although a 2-subgroup of this order-504 group has order at most eight. Thus $O_2(S) = 1$.

The product example has group order $504 \cdot 32 = 16128$, common period 36, 224 objects of each nonedge type and 4032 edges. It is Class III and faithful, but its four-type structure is not a geometry. Theorem 2 still proves that this pregeometry has a triality and no type transposition. We do not misreport the failure of flag completion as a failure of that separate symmetry conclusion.

4 A separate failure of the unrestricted chamber-orbit claim

For an odd $n \geq 3$, take a dihedron: an n-cycle embedded on the sphere with its two complementary n-gonal faces. This is a faithful thin regular map with $4n$ flags and group $D_{2n} \times C_2$. Choose adjacent reflecting generators a, b for D_{2n} and let c be the central involution swapping the faces. Then ab has order n, while bc has order two and $(ac)b = (ab)c$ has order $2n$. Consequently $H_3 = \langle b, ac \rangle = G$: there is only one Petrie polygon. The four-type structure is the original rank-three map with a universal singleton type added. It is a geometry, and G has exactly one chamber orbit, not two. This refutes source Lemma 8 in its unrestricted stated scope. The proper-parabolic hypothesis of Lemma 1 is essential. No Class III exception is claimed.

These dihedra also correct two of the source's class correspondences. For odd $n \geq 3$, their periods $n, 2, 2n$ are distinct, so they are Class I. In the four-type structure the vertex–edge incidence graph is a $2n$-cycle; both faces and the singleton Petrie object are incident with every

object of the other types. Rotation by one step of this incidence cycle interchanges vertices and edges, and fixes the remaining objects. The type-permutation quotient is exactly C_2, since the other type sizes are the distinct numbers two and one. It is not the source's trivial Class I quotient.

For even $n \geq 4$, the same dihedron has periods $n, 2, n$. It is Class II: the assignment $a \mapsto ac$, $b \mapsto b$, $c \mapsto c$ is an automorphism of $D_{2n} \times C_2$ because n is even, while the unequal period excludes the other permutations. Now H_3 has index two and is different from H_0; both are normal. Also $H_3 \cap H_2 = \langle b \rangle$. Consequently the two Petrie objects are incident with all vertices, edges and faces. The four-type structure has the same vertex–edge incidence cycle and two independent universal pairs of face and Petrie objects. It admits the independent type swaps vertex–edge and face–Petrie. Since $n > 2$ distinguishes these two pairs of type sizes, the exact type quotient is $C_2 \times C_2$, not the claimed Class II C_2.

Precise source disposition. Construction 6 defines a pregeometry for every marked group, but the source's assertion that it always is a geometry is false, even for a faithful Class III map. Lemma 7 is refuted by the torus and its Class III parallel product. Lemma 8 is false without properness of all three nonedge parabolics; its two full-type chamber orbits are proved in Lemma 1 under that exact condition. The source's universal path-residue explanation is unnecessary and false in small maps. Root's separately reviewed digonal C_2^3 example addresses Remark 9 and the claimed Class IV correlation quotient; that written companion must be acquired in full if used in a complete L023 edition. The odd and even dihedra refute the unrestricted Class I and II quotient assertions, respectively. Theorem 10's symmetry conclusion has the rigorous faithful-map replacement in Theorem 2; calling its output a geometry requires the extra completion condition. Arbitrary quotient maps with forgotten incidence flags are not silently treated as faithful maps. There is no new conjecture, Lean declaration or numerical factor.